

Department of Chemical Engineering
Indian Institute of Technology Guwahati

Course: Chemical Reaction Engineering – I
Instructor: Dr. Nageswara Rao Peela

Course code: CL208
Maximum marks: 25; Time: 2 h

Section I (7 questions of 1 Mark each)

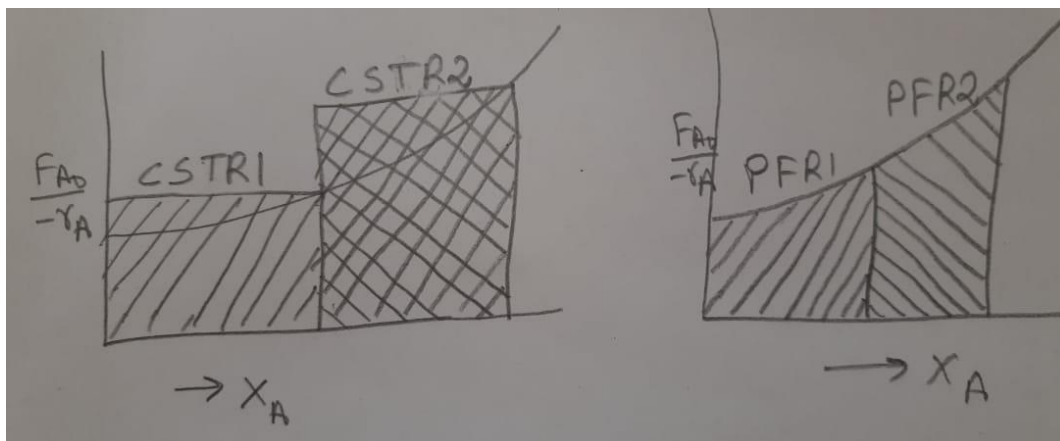
1. In batch reactor, the temporal variation is not present. True or False? **FALSE**
2. Define elementary reaction.
An elementary reaction is the one that involves a single step and the stoichiometric coefficients in this reaction are identical to the powers in the rate law.
3. If a number of Continuous Stirred Tank Reactors (CSTRs) connected in series then the performance of the system approaches to that of **PFR performance**.
4. For a set of parallel reactions, $A \rightarrow D$ and $A \rightarrow U$, define the instantaneous selectivity of D with respect to U.

$$S_{D/U} = \frac{r_D}{r_U} = \frac{\text{Rate of formation of D}}{\text{Rate of formation of U}}$$

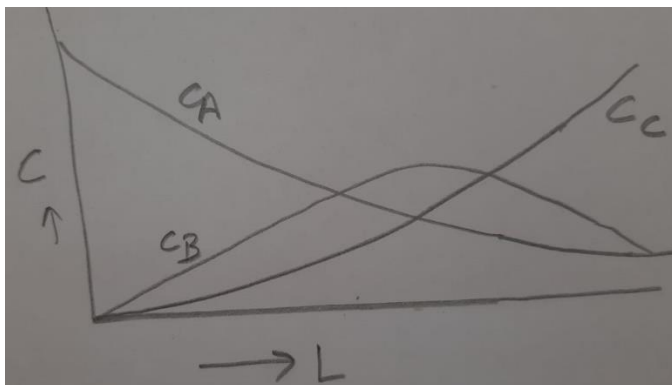
5. The molecularity is defined only for elementary reactions and it could be any fractional number. True or False? **FALSE.**
6. The reaction with larger activation energy is **more** sensitive to temperature as compared to that with smaller activation energy.
7. In the method of initial rates, a series of experiments is carried out at different **initial concentrations** and the initial rate of reaction is determined for each run.

Section II (3 questions of 2 Marks each)

1. Draw a qualitative plot between $F_{A0}/(-r_A)$ and conversion (X_A) for two CSTRs in series and two PFRs in series, separately and show the volume of the reactors as shaded area.



2. For a series reaction $A \rightarrow B \rightarrow C$, draw a qualitative plot of concentrations of A, B and C as a function of length of the reactor in a PFR. Assume all are elementary reactions with equal rate constants.



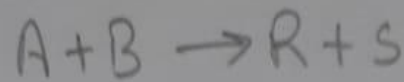
3. For a gas-phase reaction $2A + B \rightarrow 3C + 2D$ with 30% inerts, equimolar A and B are present and no C and D are present initially. calculate the volume expansion factor (ϵ) under isothermal and isobaric conditions.

$$\epsilon = \delta y_{A0} = 0.35 \times \left(\frac{3}{2} + 1 - 1 - \frac{1}{2} \right) = 0.35 \times 1 = 0.35$$

Section III (2 questions of 6 Mark each)

1. A liquid phase elementary reaction $A+B \rightarrow R+S$ is carried out in a PFR. For equimolar amounts of A and B ($C_{A0} = C_{B0} = 0.9 \text{ mol/L}$), 80% conversion is achieved. If a CSTR, 10 times as large as the PFR, is arranged in series with the existing unit, which unit needs to be arranged first (in series) to enhance the production rate?
2. The set of series reactions $A \xrightleftharpoons[k]{k} B \xrightarrow{k} C$ is conducted in a steady state isothermal, continuous stirred tank reactor (CSTR). All reactions are of first order with identical rate constants (k). The reactor volume (V) and the volumetric flow rate (v_o) are constant. The feed consists of pure A at a concentration (C_{A0}). Find out the final concentrations of A, B and C and the conversion of reactant A. Data given: $C_{A0} = 0.5 \text{ mol/L}$, $k = 1 \text{ s}^{-1}$, $V = 5 \text{ L}$; $v_o = 10 \text{ L/s}$.

SIII 1:



$$C_{A0} = C_{B0} = 0.9 \text{ mol/L}$$

$$X_A = 80\%$$

$$\begin{aligned} -r_A &= k C_A C_B = k C_A^2 \\ &= k C_{A0}^2 (1 - X_A)^2 \end{aligned}$$

PFR

$$\frac{V_P}{V_0} = \frac{X_A}{1 - X_A} \frac{1}{k C_{A0}}$$

$$V_P = \frac{0.8}{0.2} \times \frac{1}{0.9} \frac{V_0}{k}$$

$$\boxed{V_P = 4.44 \frac{V_0}{k}}$$

Two cases are possible:

Case 1: PFR followed by CSTR

Case 2: CSTR followed by PFR

Case 1: CSTR volume = $V_m = 44.4 \frac{V_0}{k}$

$$k \frac{V_m}{V_0} = \frac{X_{A2} - X_{A1}}{C_{A0} (1 - X_{A2})^2}$$

①

Incorporating all numbers and simplifying gives

$$40X_{A_2}^2 - 81X_{A_2} + 40.8 = 0$$

on solving: $X_{A_2} = 0.94$

case 2: CSTR followed by PFR

$$k \frac{V_m}{v_0} = \frac{X_{A_1}}{C_{A_0}(1-X_{A_1})^2}$$

$$V_m = 44.4 \frac{v_0}{k}$$

$$C_{A_0} = 0.9 \text{ mol/L}$$

$$40(1-X_{A_1})^2 = X_{A_1}$$

$$X_{A_1} = 0.85$$

PFR:

$$4.44 \frac{v_0}{k} \frac{1}{v_0 C_{A_0}} = \frac{1}{k C_{A_0}^2} \left(\frac{1}{1-X_{A_2}} - \frac{1}{1-X_{A_1}} \right)$$

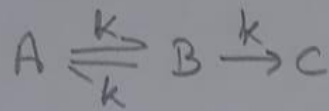
$$\therefore X_{A_2} = 0.91$$

Case 1 gives higher overall conversion

$\therefore$ The reactor arrangement should be PFR followed by CSTR.

(2)

S III 2



$$C_{A0} = 0.5 \text{ mol/L}; k = 1 \text{ s}^{-1}; V = 5 \text{ L}; v_0 = 10 \text{ L/s}$$

$$-r_A = kC_A - kC_B$$

$$-r_B = kC_B + kC_B - kC_A$$

$$r_C = kC_B$$

MB on CSTR:

for A: $v_0 C_{A0} = v_0 C_A + (-r_A)V$

$$v_0 C_{A0} = v_0 C_A + kC_A V - kC_B V$$

$$C_{A0} = C_A + \frac{kV}{v_0} C_A - \frac{kV}{v_0} C_B$$

$$C_{A0} = \left(1 + \frac{kV}{v_0}\right) C_A - \frac{kV}{v_0} C_B \quad \text{--- (1)}$$

for B: $v_0 C_{B0} = v_0 C_B + (-r_B)V$

$$0 = \left(1 + \frac{2kV}{v_0}\right) C_B - \frac{kV}{v_0} C_A \quad \text{--- (2)}$$

putting values in eq. (1) & (2) & simplifying gives

$$3C_A - C_B = 1 \quad \text{--- (3)}$$

$$-0.25C_A + C_B = 0 \quad \text{--- (4)}$$

(3)

Solving eq.s (3) & (4) gives

$$C_A = 0.364 \text{ mol/L}$$

$$C_B = 0.09 \text{ mol/L}$$

$$C_C = C_{A0} - C_A - C_B \quad (\text{for liquid phase reactions})$$

$$C_C = 0.045 \text{ mol/L}$$

$$X_A = \frac{C_{A0} - C_A}{C_{A0}}$$
$$= 0.272$$

$\therefore$ The conversion of A = 27.2 %